of flowering plants, 317 species of mammals, 969 species of birds, 389 species of reptiles, 206 species of amphibians, and so on. In any *sustainable agricultural system*, the concern regarding bio-diversity cannot be ignored.

4.4 WATER RESOURCES

Water is the most important source of energy in the Indian economy. About 25 per cent of electricity generated in the economy is from the hydel sources. The other important use of water is in irrigation. In a country where agriculture anchors the economy, availability of water can make all the difference; it can either stimulate the economic activity or depress it altogether.

The important sources of water can be classified into: (*i*) surface water, and (*ii*) ground water. *Surface water* is available from such sources as rivers, lakes, etc. *Ground water* is available from wells, springs, etc. Other sources of water which have not as yet been tapped in the country, but nevertheless represent a potential source are: wastewater, saline lakes, saline springs, etc. Surface water sources are replenished by rainfall.

Of the two sources, surface water is more important and possesses potential for growth in future. Surface water is available in the form of vast network of rivers available in the country.

Overall, India possesses large reservoirs of water, but these are inadequate as compared to country's requirements. Compared to countries such as the USA, which stores about 5,000 cubic meters per capita and China, which stores around 1,000 cubic meters per capita, India's dams can only store 200 cubic meters per person.

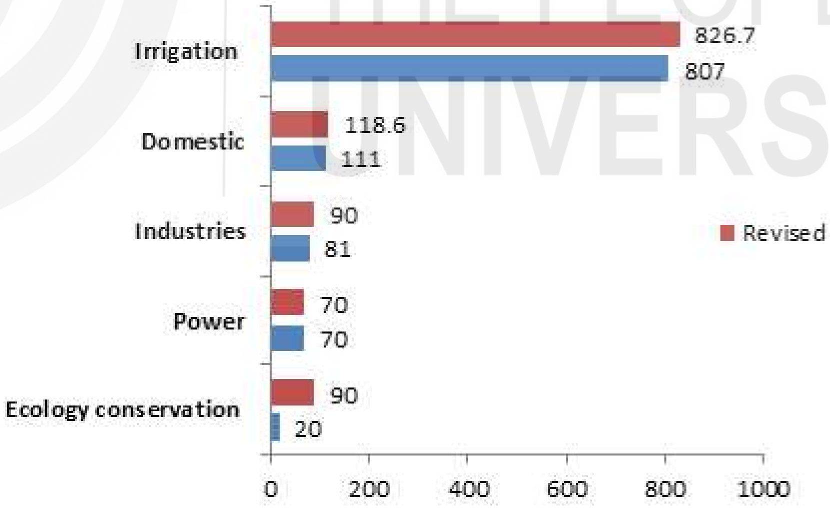

Fig. 4.2: Estimates of Demand for Water in India

Through the decades, profligacy in the use of water by industrial units and during irrigation has brought the country close to a "water famine". In fact, many parts of India are already water-stressed, with per capita availability significantly less than the national average. At about 1,000 cubic meters, the national average, too, falls way short of the 'comfort point' set by the World Bank. No wonder many states remain at daggers drawn over sharing water from rivers that flow through them.

4.4.1 Water Issues and Solutions

The principal issues facing the country are as follows: (i) Demand for water is increasing from all sectors, (ii) Lack of a rational water pricing policy between and within sectors is further driving demand, (iii) Policies and institutions mandated to solve conflicts are directly or indirectly contributing to further conflicts, (iv) New conflicts are increasingly arising within states rather than between states, and (v) Conflicts over groundwater are widespread across the country.

Solutions

The following solutions suggest themselves:

1) The one over-riding lesson from the global resolution in the provision of public services is that competition matters. Hence, it is important to unbundle the bulk provider of the irrigation system from the distribution function. Further, there should be variety of forms– cooperatives, and private sector players– to handle water distribution to farmers.
2) There is a need to increase user charges for the services provided, as well as to increase budgetary support. But user charges or tariffs linked to costs can only be possible if services are provided in an efficient and accountable manner.
3) It is by ensuring the economic and social development of local communities and their participation that India can hope to build major dams.
4) The government should give formal water entitlements to people as with land and other property rights. Once established, such entitlements give rise to a fundamental and healthy changes including improvement in efficiency of resource use.

Four interconnected programmes as follows assume significance in this context: (i) watershed development programme, (ii) renovation of all water bodies linked to agriculture which have fallen into disuse, (iii) correcting the deterioration of public irrigation works, notably state canal systems, due to cumulative neglect of maintenance and repair over the years, and (iv) rainwater harvesting.

Another encouraging development is that more and more futuristic technologies are being readied in research labs to make desalination and water treatment processes more energy efficient.

4.4.2 National Law on Water

Why is a national law on water necessary?

1) Under the Indian Constitution water is primarily a State subject, but it is an increasingly important national concern in the context of:
   a) the judicial recognition of the right to water as a part of the fundamental right to life;
   b) the general perception of an imminent water crisis, and the dire and urgent need to conserve this scarce and precious resource;
   c) the severe and intractable inter-use and inter-State conflicts;

d) the pollution of rivers and other water sources, turning rivers into sewers or poison and contaminating aquifers;

e) the long-term environmental, ecological and social implications of projects to augment the availability of water for human use;

f) the equity implications of the distribution, use and control of water;

g) the international dimensions of some of India's rivers; and

h) the emerging concerns about the impact of climate change on water and the need for appropriate responses at local, national, regional, and global levels.

It is clear that the above considerations cast several responsibilities on the Central government, apart from those of the State governments. Given these and other concerns, the need for an overarching national water law is self-evident.

2) Several States are enacting laws on water and related issues. These can be quite divergent in their perceptions of and approaches to water. Some divergences from State to State may be inevitable and acceptable, but extreme and fundamental divergences will create a very muddled situation. A broad national consensus on certain basics seems very desirable.

3) Different State governments tend to adopt different legal positions on their rights over the waters of a river basin that straddles more than one State. Such legal divergences tend to render the resolution of inter-State river-water conflicts extremely difficult. A national statement of the general legal position and principles that should govern such cases seems desirable.

4) Water is one of the most basic requirements for life. If national laws are considered necessary on subjects such as the environment, forests, wildlife, biological diversity, etc., a national law on water is even more necessary. Water is as basic as (if not more basic than) those subjects.

5) Finally, the idea of a national water law is not something unusual or unprecedented. Many countries in the world have national water laws or codes, and some of them (for instance, the South African National Water Act of 1998) are widely regarded as very enlightened. The considerations behind those national codes or laws are relevant to India as well, although the form of a water law for India will clearly have to be guided by the nature of the Indian Constitution and the specific needs and circumstances of this country.

Check Your Progress 1

1) Outline the significance of the size of land area and its location for the Indian economy.

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